Learnix — Project Report

EXPLAINABLE COURSE RECOMMENDER
SYSTEM WORKFLOW & DESIGN

FRONTEND
Next.js 16 (App Router) · Turbopack · React 19 · TypeScript · Tailwind CSS v4 · Framer Motion · Zustand · Lucide Icons

BACKEND & DATA
Server Actions · Route Handlers · Prisma ORM 6 · PostgreSQL · Supabase (pooled)

AUTH & VALIDATION
Auth.js / NextAuth v5 · JWT sessions · bcryptjs · Zod schemas

TOOLING & TESTING
pnpm · ESLint · Vitest · Testing Library · Prisma Studio

END-TO-END WORKFLOW

1. **Sign up / log in.** Auth.js (NextAuth v5) Credentials provider; password hashed with bcrypt; session kept as a signed JWT, verified at the edge by middleware before any protected route renders.
2. **Onboarding wizard (cold start).** Four steps capture: target *career* (sets the skill profile to match against), *interests* (keyword boost), *transcript* (courses already completed), and *limits* (weekly study-hour budget + unavailable days). This seeds a usable profile before any click-history exists.
3. **Profile → skill vector.** Confirmed transcript courses are summed into a 5-axis "current skills" vector (ML, Statistics, Systems, Algorithms, Engineering practice), capped at 1 per axis.
4. **Scoring engine runs per request.** Every catalogue course is scored live against the student's skill vector, chosen career target, and five weighted signals — never pre-computed or cached per-student.
5. **Rank, explain, exclude.** Hidden courses and courses on a declared day-off are removed *before* ranking; the rest are sorted by match % and annotated with a plain-language "why it fits" built from the top contributing signals.
6. **Home / Explore / Course detail.** Home shows the ranked top matches; Explore adds full search, department/day/workload filters and alternate sort modes; the course page shows the full score breakdown, instructor, reviews and syllabus.
7. **Save / Hide (implicit feedback).** One-click actions persist per student and immediately invalidate the server cache, so the next render reflects the change with no stale data.
8. **Compare & Path.** Compare lines up saved courses side by side with clash detection (shared day + overlapping time). Path extends the scorer into a greedy multi-term planner — topologically ordered by prerequisite, packed to the student's credit/hour budget.
9. **Progress.** Degree-requirement completion and GPA are *derived live* from the transcript on every request — never stored — so they can't go stale or be hand-edited.
10. **Settings (steering).** Five signal-weight sliders let the student re-rank the entire catalogue instantly; notification and accessibility preferences live here too.
11. **Updates.** Deadline, seat-availability and new-course notifications, with per-student read state.

RECOMMENDER-SYSTEM CONCEPTS APPLIED

CONTENT-BASED FILTERING

Every course and every career target is represented as a **feature vector** in the same 5-dimensional skill space — the textbook content-based architecture: item profile and user-target profile compared in one shared space, with no reliance on other users' behaviour.

WEIGHTED SIMILARITY SCORING

The *goals* signal is a weighted dot-product (cosine-style) similarity between a course's skill vector and the career target, normalised to 0–99.

```
goals = 38 + 105 × (Σ sk·target / Σ target) + interest bonus
```

NEEDS-BASED / GAP-DRIVEN RANKING

The *gap* signal re-weights that same similarity by how far the student's **current** skills fall short of the target — a course that closes a bigger deficit outranks one that only repeats a skill already held. This is a needs-driven personalisation strategy distinct from pure similarity matching.

MULTI-CRITERIA HYBRID RANKING

Five independent signals — goals, skill-gap, past-course history, schedule fit, credit value — are combined by a **weighted-sum multi-attribute utility function**, normalised so weights never need to sum to 100:

```
match % = Σ(subscore[k] × weight[k]) / Σ weight[k]
```



Goals 30% · Gap 25% · History 20% · Schedule 15% · Cost 10%

KNOWLEDGE / CONSTRAINT-BASED FILTERING

Prerequisites, weekly-hour budget and day-off conflicts are treated as **hard knowledge constraints** — a course either gets excluded outright (hidden, day clash) or has its subscore discounted (unmet prerequisite, oversized workload), rather than being silently buried by a low score.

EXPLAINABILITY

Every match percentage decomposes into its per-signal contributions, and the top three drive a plain-language reason shown with the course — directly addressing the "black box" critique levelled at pure similarity-based recommenders.

INTERACTIVE / STEERABLE RECOMMENDATIONS

Weight sliders in Settings let the student re-rank the whole catalogue live — a critique-based steering loop, letting explicit preference feedback override the defaults instantly.

COLD-START HANDLING

The onboarding wizard builds a usable profile (career, interests, prior transcript) **before** any interaction history exists, solving the new-user cold-start problem that pure collaborative methods cannot.